

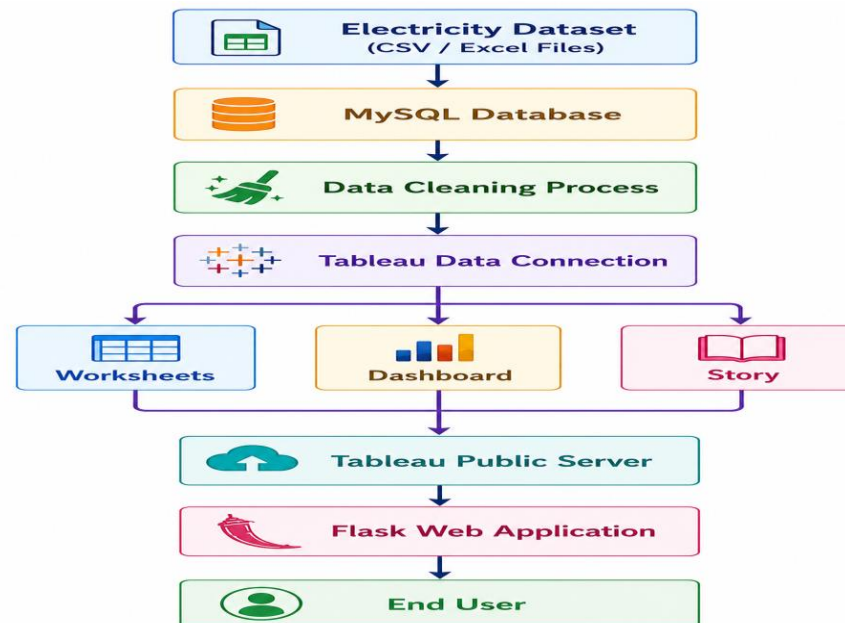
Project Design Phase-II

Data Flow Diagram & User Stories

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data Flow Diagrams:

The DFD represents the complete flow of electricity consumption data from data collection and processing to visualization in Tableau dashboards and final presentation through Tableau Public and Flask web application..



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Power Grid Analyst (Internal)	Data Ingestion & Upload	USN-1	As a grid analyst, I want to upload massive daily electricity consumption CSV logs into the dashboard pipeline.	System handles over 16,800 rows without format errors or memory timeouts.	High	Sprint-1
	Data Segmentation	USN-2	As a grid analyst, I want the system to categorize consumption automatically by 5 national zones (North, West, South, East, NE).	Data automatically segments into proper regional buckets during processing.	High	Sprint-1
	Regional Trend Optimization	USN-3	As a grid analyst, I want to apply Context Filters so that when I filter a region, the top/bottom state rankings update accurately.	I State ranking charts update instantly in the correct sequence without dynamic calculation glitches.	High	Sprint-1
Director / Policy Maker (Web User)	Baseline Comparison View	USN-04	As a director, I want to see a side-by-side comparison of 2019 baseline vs 2020 lockdown usage.	A dual-year synchronized filled map visualizes the drop percentages instantly on hover.	Medium	Sprint-2
	Analytical Insights Export	USN-05	As a policy maker, I want to export the cleaned cross-filtered charts as a PDF summary report.	Exported report retains the current dashboard filter settings	Medium	Sprint-2
System Administrator	Pipeline Maintenance	USN-06	As an admin, I want to schedule automatic updates of energy data from regional grid servers every midnight.	In-memory hyper extracts refresh smoothly without interrupting analyst active sessions.	Low	Sprint-2